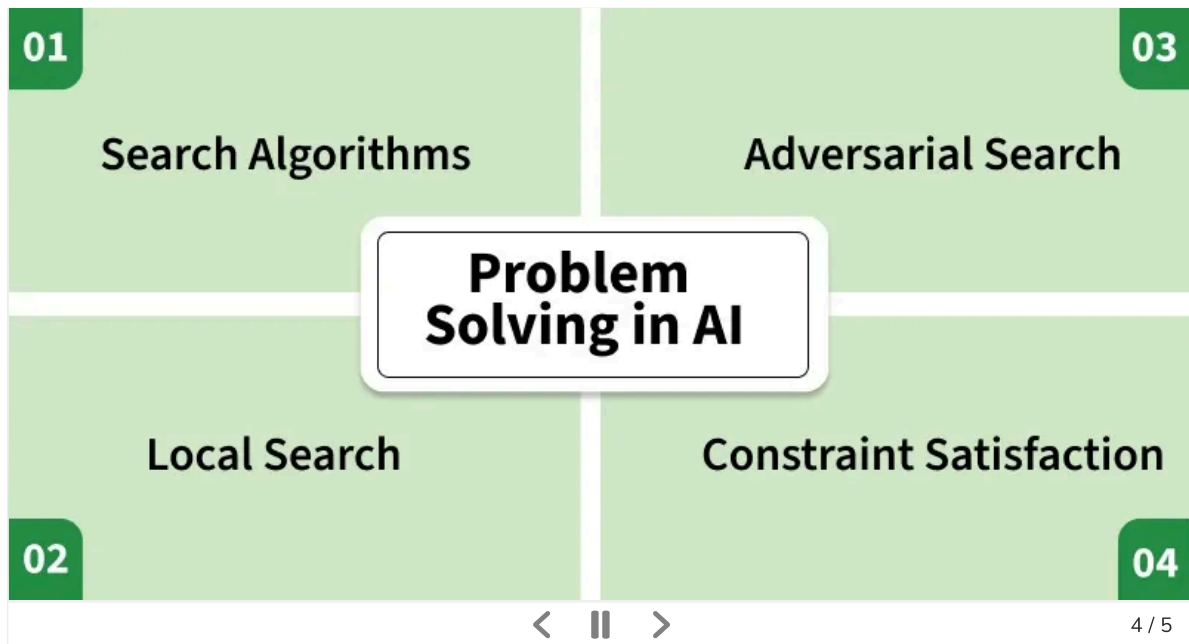




Artificial Intelligence Tutorial | AI Tutorial

Last Updated : 26 Sep, 2025

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines which helps in allowing them to think and act like humans. It involves creating algorithms and systems that can perform tasks which requiring human abilities such as visual perception, speech recognition, decision-making and language translation.



Types of Artificial Intelligence

Artificial Intelligence (AI) is classified into:

- [Types of AI Based on Capabilities](#)
- [Types of AI Based on Functionalities](#)

What is an AI Agent?

An AI agent is a software or hardware entity that performs actions autonomously with the goal of achieving specific objectives.

- [AI agent](#)
- [types of AI Agents](#)

Problem Solving in AI

Problem-solving is a fundamental aspect of AI which involves the design and application of algorithms to solve complex problems systematically.

1. Search Algorithms in AI

Search algorithms navigate through problem spaces to find solutions.

- [Search algorithms](#)
- [Breadth-First Search \(BFS\)](#)
- [Depth-First Search \(DFS\)](#)
- [Uniform Cost Search \(UCS\)](#)
- [Bidirectional search](#)
- [Greedy Best-First Search](#)
- [A Search* Algorithm](#)

2. Local Search Algorithms

Local search algorithms operates on a single current state (or a small set of states) and attempt to improve it incrementally by exploring neighboring states.

- [Local search algorithms](#)
- [Hill-Climbing Search Algorithm](#)
- [Local Beam Search](#)

3. Adversarial Search in AI

Adversarial search deal with competitive environments where multiple agents (often two) are in direct competition with one another such as in games like chess, tic-tac-toe or Go.

- [Adversarial search](#)
- [Minimax Algorithm](#)
- [Alpha-Beta Pruning](#)

4. Constraint Satisfaction Problems

Constraint Satisfaction Problem (CSP) is a problem-solving framework that involves variables each with a domain of possible values and constraints limiting the combinations of variable values.

- [Constraint Satisfaction Problem \(CSP\)](#)
- [Constraint Propagation in CSP's](#)
- [Backtracking Search for CSP's](#)

Knowledge, Reasoning and Planning in AI

Knowledge representation in Artificial Intelligence (AI) refers to the way information, knowledge and data are structured, stored and used by AI systems to reason, learn and make decisions.

Common techniques for knowledge representation include:

- [Knowledge representation in Artificial Intelligence \(AI\)](#)
- [Semantic Networks](#)
- [Frames](#)
- [Ontologies](#)
- [Logical Representation](#)

First Order Logic in Artificial Intelligence

First Order Logic (FOL) is use to represent knowledge and reason about the world. It allows for the expression of more complex statements involving objects, their properties and the relationships between them.

- [First Order Logic \(FOL\)](#)
- [Knowledge Representation in First Order Logic](#)
- [Syntax and Semantics of First Order Logic](#)
- [Inference Rules in First Order Logic](#)

Reasoning in Artificial Intelligence

Reasoning in Artificial Intelligence (AI) is the process by which AI systems draw conclusions, make decisions or infer new knowledge from existing information.

Types of reasoning used in AI are:

- [Reasoning in Artificial Intelligence \(AI\)](#)
- [Types of Reasoning in AI](#)
- [Deductive Reasoning](#)
- [Inductive Reasoning](#)
- [Abductive Reasoning](#)
- [Fuzzy Reasoning](#)

Planning in AI

Planning in AI generates a sequence of actions that an intelligent agent needs to execute to achieve specific goals or objectives. Some of the planning techniques in artificial intelligence includes:

- [Planning in AI](#)
- [Forward State Space Search](#)
- [Markov Decision Processes \(MDPs\)](#)
- [Hierarchical State Space Search \(HSSS\)](#)

Uncertain Knowledge and Reasoning

Uncertain Knowledge and Reasoning in AI refers to the methods and techniques used to handle situations where information is incomplete, ambiguous or uncertain. For managing uncertainty in AI following methods are used:

- [Uncertain Knowledge and Reasoning in AI](#)
- [Dempster-Shafer Theory](#)
- [Probabilistic Reasoning](#)
- [Fuzzy Logic](#)
- [Neural Networks with dropout](#)

Types of Learning in AI

Learning in Artificial Intelligence (AI) refers to the process by which a system improves its performance on a task over time through experience, data or interaction with the environment.

1. Supervised Learning

In Supervised Learning model are trained on labeled dataset to learn the mapping from inputs to outputs.

Various algorithms are:

- [Supervised Learning](#)
- [Linear Regression](#)

- [Logistic Regression](#)
- [Decision Trees](#)
- [Support Vector Machines \(SVM\)](#)
- [k-Nearest Neighbors](#)
- [Naïve Bayes](#)
- [Random Forests](#)

2. Semi-supervised learning

In Semi-supervised learning the model uses both labeled and unlabeled data to improve learning accuracy.

- [Semi-supervised learning](#)

3. Unsupervised Learning

In Unsupervised Learning the model is trained on unlabeled dataset to discover patterns or structures.

- [Unsupervised Learning](#)
- [K-Means Clustering](#)
- [Principal Component Analysis \(PCA\)](#)
- [Hierarchical Clustering](#)
- [DBSCAN \(Density-Based Spatial Clustering of Applications with Noise\)](#)

4. Reinforcement Learning

In Reinforcement Learning the agent learns through interactions with an environment using feedbacks.

- [Reinforcement Learning](#)
- [Q-Learning](#)
- [Deep Q-Networks \(DQN\)](#)
- [Markov decision processes \(MDPs\)](#)
- [Bellman equation](#)

5. Deep Learning

Deep Learning focuses on using neural networks with many layers to model and understand complex patterns and representations in large datasets.

- [Deep Learning](#)
- [Neurons](#)
- [Single Layer Perceptron](#)
- [Multi-Layer Perceptron](#)
- [Artificial Neural Networks \(ANNs\)](#)
- [Feedforward Neural Networks \(FNN\)](#)
- [Convolutional Neural Networks \(CNN\)](#)
- [Recurrent Neural Networks \(RNNs\)](#)
- [Long Short-Term Memory \(LSTM\) networks](#)
- [Gated Recurrent Units Networks \(GRU\)](#)

Probabilistic models

Probabilistic models in AI deals with uncertainty making predictions and modeling complex systems where uncertainty and variability play an important role. These models help in reasoning, decision-making and learning from data.

- [Probabilistic models](#)
- [Naive Bayes Classifier](#)
- [Monte Carlo Methods](#)
- [Expectation-Maximization \(EM\) Algorithm](#)

Communication, Perceiving and Acting in AI and Robotics

Communication in AI and robotics helps in the interaction between machines and their environments which uses natural language processing. Perceiving helps machines using sensors and cameras to interpret their surroundings accurately. Acting in robotics includes making informed decisions and performing tasks based on processed data.

1. [Natural Language Processing \(NLP\)](#)

- [Speech Recognition](#)
- [Natural Language Generation](#)
- [Chatbots](#)
- [Machine Translation](#)

2. [Computer Vision](#)

- [Image Recognition](#)
- [Facial Recognition](#)
- [Optical Character Recognition](#)

3. [Robotics](#)

Generative AI

Generative AI focuses on creating new data examples that resemble real data, effectively learning the distribution of data to generate similar but distinct outputs.

- [Large Language Models](#)
- [GPT \(Generative Pre-trained Transformer\)](#)
- [BERT \(Bidirectional Encoder Representations from Transformers\)](#)
- [T5 \(Text-to-Text Transfer Transformer\)](#)
- [Conditional GAN \(cGAN\)](#)
- [CycleGAN](#)
- [Style GANs](#)

We've covered the AI tutorial which is important for developing intelligent systems and helps in making the perfect balance of simplicity and capability.

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